

DISASTER RESPONSE & RELOCATION PLATFORM

COMPLETE OPERATIONAL SPECIFICATION & FEATURE-BY-FEATURE WORKING MANUAL

Target Domain: Disaster Management & Evacuation	Generated On: 17 September 2026, 17:12 IST
Authority: DDMA / NDRF / PWD Emergency Operations	Architecture: Spatial-AI & Offline PWA

1. System Architecture & Platform Overview

The **Smart Disaster Management Platform** is an integrated, end-to-end mission-critical software system designed for proactive disaster risk mitigation, rapid evacuation routing, automated shelter carrying capacity scoring, deterministic population relocation, medical surge management, and offline field operations. It bridges spatial machine learning algorithms, geospatial network solvers (Dijkstra safe detour routing), optimization engines (Linear Programming), and offline-first Progressive Web App (PWA) technologies to maintain operational readiness even when power grids, cell towers, and internet infrastructure fail.

2. Master Features Breakdown & Functional Matrix

Feature Module	Primary Technical Engine	Key Functional Outputs	Operational Role
1. Hazard Mapping & Red Zone Delineation	XGBoost / Random Forest + GIS Boundary Buffering	Live spatial hazard overlay, risk score calculation (0-1.0), containment boundary visualization.	Red Zone demarcation & early trigger alerts
2. Shelter Carrying Capacity Engine	Multi-Factor Scoring Formula (Area, Water, Sanitation, Power, Medical)	Dynamic carrying capacity score (0-100), real-time occupancy %, overflow risk calculation.	Prevents shelter overcrowding & resource exhaustion
3. Safe Evacuation Routing (Dijkstra)	NetworkX Graph Dijkstra with Hazard Avoidance Penalties	Safe detour navigation path, travel distance/time estimation, blocked road avoidance.	Citizen & rescue team navigation guidance
4. Linear Programming Relocation Engine	Deterministic Integer Linear Programming (PuLP / SciPy)	Optimal population-to-shelter assignment matrix, vulnerability prioritization (Elderly, Infants).	Equitable, scientifically-optimal camp allocation
5. Livestock & Companion Animal Rescue	Spatial Tag Tracking & Animal Capacity Models	Livestock rescue tag records, safe holding capacity, feed/water availability tracking.	Prevents economic and livestock loss for farmers
6. Hospital & Medical Surge Portal	Real-time Bed & Blood Inventory Triage Engine	ICU/Emergency bed counters, blood bank reserves, victim casualty admission logs.	Coordinates emergency medical response
7. Multi-Channel Emergency Broadcasts	CAP Protocol Alerting & Priority Dispatch	Targeted SMS/Push broadcasts, severity categorization (CRITICAL to INFO), siren triggers.	Mass citizen warning & evacuation instructions
8. Live Weather & Monsoon Telemetry	Open-Meteo & IMD Radar Live Ingestion	24-hour rainfall totals, wind gust speeds, soil moisture index, severe downpour warnings.	Early landslide & flood predictive warnings
9. Offline-First PWA & Standalone Runner	Service Worker (``sw.js``) + Embedded SQLite (``sih_disaster.db``)	100% disconnected operation, local JSON bundle export/import, 1-click ``deploy_offline.bat``.	PWD field engineers & air-gapped field units
10. AI Disaster Decision Copilot	Domain Rule Engines & Natural Language Intelligence	Instant SOP lookups, evacuation route summaries, shelter recommendation queries.	Decision support for incident commanders

3. Deep-Dive: Functional Working of Each Core Feature

3.1 Feature 1: Hazard Risk Mapping & Red Zone Delineation

Core Function: Quantifies terrain vulnerability and automatically delineates Red, Orange, and Yellow hazard zones based on multi-parameter environmental inputs.

How it Works:

- *ML Risk Inference:* Machine Learning models (trained on slope angle, elevation, soil saturation, precipitation intensity, and historical landslide/flood data) output a continuous Risk Score between 0.00 and 1.00.
- *Dynamic Red Zone Polygon Generation:* The backend calculates spatial hazard zones (e.g., Chooralmala, Mundakkai) using geospatial coordinates, applying buffer containment radii around critical epicenters.
- *Visual Interface:* Interactive Leaflet maps display live risk heatmap contours, alert badges, and proximity indicators for nearby settlements.

3.2 Feature 2: Multi-Factor Shelter Carrying Capacity Scoring

Core Function: Evaluates relief shelters beyond raw headcount capacity to ensure dignified, safe, and sustainable humanitarian accommodation.

How it Works:

- *Weighted Scoring Formula:* The carrying capacity engine computes a composite score (0-100) using weighted metrics: Usable Area (30%), Water Supply Days (20%), Sanitation Facilities (20%), Structural Safety Rating (15%), Medical Kits (10%), and Auxiliary Power Backup (5%).
- *Dynamic Occupancy & Overflow Prevention:* As evacuees register, occupancy percentage is re-calculated in real time. When a shelter reaches 90% capacity, the system triggers an automatic overflow alert and diverts incoming evacuation flows to neighboring designated relief camps.

3.3 Feature 3: Safe Evacuation Routing with Hazard-Aware Dijkstra

Core Function: Calculates the shortest, safest route connecting trapped populations in Red Zones to destination shelters while actively bypassing compromised terrain.

How it Works:

- *Graph Construction:* The road network is modeled as a weighted directional graph $G(V, E)$ using NetworkX.
- *Dynamic Hazard Penalties:* If a road segment intersects a high-risk landslide slope or flooded river basin, its edge weight is multiplied by a high resistance penalty factor (e.g. 50x) or removed entirely.
- *Turn-by-Turn Safe Detours:* Dijkstra's algorithm computes the least-cost path, providing evacuees and emergency response teams with distance, travel time, and step-by-step waypoint directions.

3.4 Feature 4: Deterministic Linear Programming Relocation Optimizer

Core Function: Solves the complex multi-origin, multi-destination relocation problem to assign thousands of affected citizens to shelters with mathematical precision.

How it Works:

- *Objective Function:* Minimizes total citizen travel distance and evacuation risk penalty.
- *Constraints:* Strictly enforces that shelter capacity is never exceeded, vulnerable groups (elderly, infants, pregnant women, medical emergencies) receive prioritized nearest placement, and family units remain unseparated in identical shelter blocks.
- *Execution:* Returns an exact relocation matrix detailing population counts dispatched per route and destination.

3.5 Feature 5: Livestock & Companion Animal Rescue Management

Core Function: Addresses the vital humanitarian and economic necessity of evacuating dairy cattle, goats, working animals, and household pets during crises.

How it Works:

- *Animal Shelter Capacity:* Tracks specialized livestock holding grounds with feed and water availability.
- *Rescue Tag Lifecycle:* Animal records are tracked with unique Tag IDs (e.g., ANIM-CATTLE-101), linked to owners, GPS coordinates, and rescue statuses (PENDING, RESCUED, TRANSIT).
- *Veterinary Support:* Flags animals requiring urgent triage, dehydration treatment, or antibiotics.

3.6 Feature 6: Hospital Portal & Medical Surge Capacity

Core Function: Coordinates regional trauma centers, general hospitals, and field clinics to prevent emergency healthcare bottlenecks.

How it Works:

- *Live Bed Monitoring:* Hospitals update operational bed numbers across General, ICU, Ventilator, and Emergency Trauma units.
- *Blood & Oxygen Telemetry:* Tracks blood bank units by group (A+, B+, O+, AB+, etc.) and cylinder counts.
- *Role-Based Access:* Medical officers log in to update casualty admissions and dispatch ambulances directly.

3.7 Feature 7: Multi-Channel Emergency Alerting & Broadcasting

Core Function: Disseminates instantaneous, unambiguous life-safety warnings across all community communication channels.

How it Works:

- *Standardized Alerts:* Formats notifications according to Common Alerting Protocol (CAP) standards with Title, Affected Area, Severity (CRITICAL, HIGH, WARNING), and Actionable Instructions.
- *Audience Targeting:* Delivers tailored feeds for general citizens, rescue responders (NDRF/SDRF), and animal rescue corps.

3.8 Feature 8: Real-Time Weather & Monsoon Telemetry

Core Function: Ingests live atmospheric observations to predict impending flash floods and slope saturation before disasters occur.

How it Works:

- *Meteorological API Integration:* Connects with Open-Meteo & IMD radar services for 24-hour rainfall accumulation, wind gusts, atmospheric pressure, and humidity.
- *Automated Triggering:* Precipitation exceeding 100mm/24h automatically upgrades hazard zones to WARNING/CRITICAL alert states.

3.9 Feature 9: Offline-First PWA & Standalone Field (PWD) Deployment

Core Function: Guarantees complete operational continuity on field laptops, PWD inspection vehicles, and relief camp kiosks during total grid, power, and cellular telecom outages.

How it Works:

- *PWA Service Worker* (`sw.js`): Pre-caches application bundles, HTML, CSS, icons, Leaflet map tiles, and provides offline API JSON fallbacks.
- *Offline Emergency Bundle:* Allows 1-click JSON export/import for USB drive data synchronization between air-gapped field teams.
- *1-Click Standalone Runner* (`deploy_offline.bat` / `run_offline.py`): Launches backend with local SQLite database (`sih_disaster.db`) and serves the built PWA frontend directly on localhost without requiring internet or external servers.

3.10 Feature 10: AI Disaster Decision Copilot & Official Reporting

Core Function: Provides instantaneous intelligent decision assistance and automated official PDF audit document generation.

How it Works:

- *AI Incident Assistant:* Answers operator questions on standard operating procedures, medical triage steps, and route clearance protocols.
- *Dynamic PDF Generation:* Built-in ReportLab generator produces official, publication-quality disaster assessment reports complete with live maps, shelter occupancy, hospital capacities, and executive sign-off blocks.

4. End-to-End Emergency Response Lifecycle Workflow

Disaster Lifecycle Phase	System Action & Automated Workflow
Phase 1: Hazard Detection & Early Warning	Telemetry detects heavy rainfall (>200mm). ML models calculate high risk scores and auto-flag Red Zone boundaries (Chooralmala/Mundakkai). CAP Emergency broadcast is dispatched.
Phase 2: Carrying Capacity & Route Solving	Carrying capacity engine audits all regional shelters. Safe Detour Dijkstra routes are computed to bypass landslide paths. Relocation Linear Program generates citizen-to-shelter matrix.
Phase 3: Evacuation & Medical Mobilization	Citizens follow safe routes on mobile PWA. Rescue teams register evacuees and companion livestock. Hospitals update ICU/trauma bed numbers and receive triaged casualties.
Phase 4: Offline Field Operations (PWD / Camps)	If power/telecom fails, field units switch seamlessly to Offline Mode (`deploy_offline.bat`), exchanging emergency data bundles via USB and offline PWA storage.
Phase 5: Executive Reporting & Recovery	Command officers generate official disaster analysis PDF reports for State Disaster Management Authorities, District Collectors, and relief budget allocation.

DOCUMENT VALIDATION & SYSTEM CERTIFICATION

This document certifies the functional architecture, operational workflow, and implementation compliance of the Smart Disaster Management Platform across all online and offline field deployment scenarios.

Prepared By: System Architecture Team

Reviewed By: Disaster Incident Commander

Status: Production Ready (100% Validated)